

EECE 571Q

Lecture 11: Do you believe in magic?

Tuesday 16 June 2026

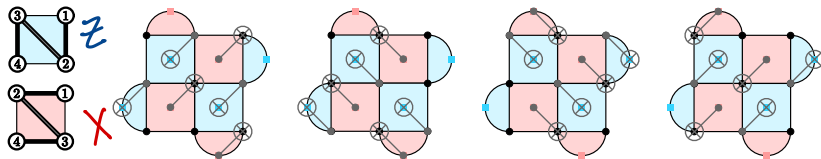
Announcements

- Quiz 5 today
- Exam reference sheet due Thursday 18 June 23:59

Last time

We discussed how errors manifest on the surface code, and some considerations for dealing with them.

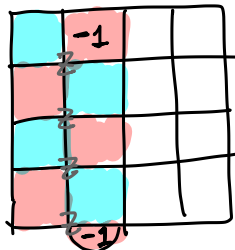
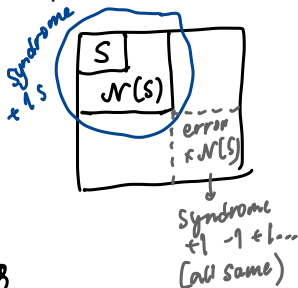
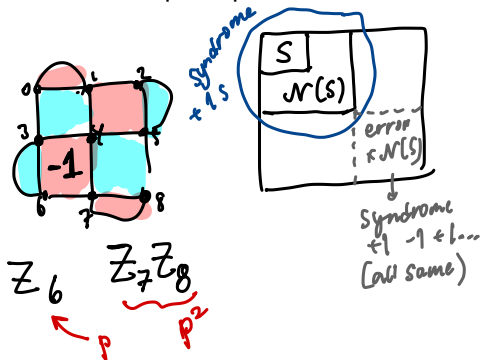
Example: gate scheduling in syndrome measurement



Last time

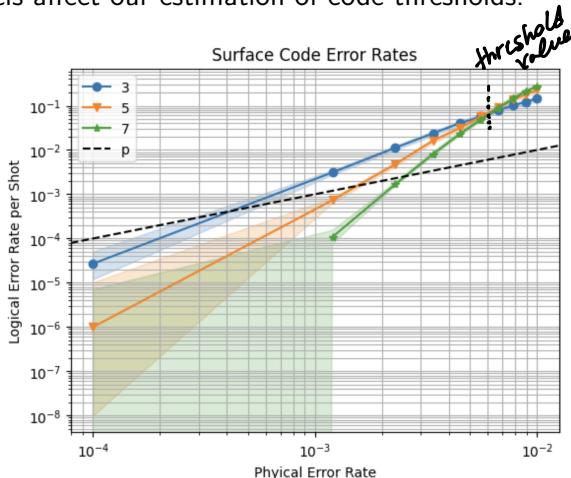
We discussed decoding at a high level:

- maximum-likelihood decoding (choosing coset representatives for corrective operations)
- minimum-weight perfect matching (find error chains that match up endpoints in surface code syndrome measurements)



Last time

We ran simulations with stim to explore how code distance and error models affect our estimation of code thresholds.



Learning outcomes

- 1 Explain why at least one non-transversal gate is necessary for universal FTQC
- 2 Define magic states, and inject them to apply operations with gate teleportation
- 3 Perform magic state distillation with the $[[15, 1, 3]]$ quantum Reed-Muller code

Transversal gates

Transversal gates naturally have fault-tolerant properties.
Different codes admit different transversal gates.

Example: transversal gates in Steane code.

$$H^{\otimes 7} (ZZZZIIII) H^{\otimes 7} = XXXXIIII$$

$$ZZIIIZZI$$

$$ZIZIZIZI$$

$$H^{\otimes 7} (XXXXIIII) H^{\otimes 7} = ZZZZIIII$$

$$XXIIXXI$$

$$XIIXIXI$$

$$\wedge ZZZZZZ$$

Is H transversal?

Is $CNOT$ transversal?

Is S transversal?

$$S^{\otimes 7} (XXXXIIII) S^{\otimes 7} = YYYYYI$$

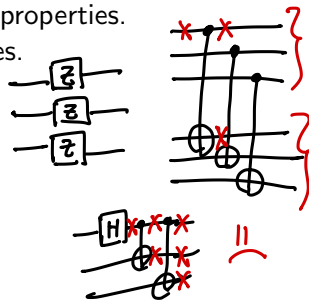
$$S^{\otimes 7} (ZZZZIIII) S^{\otimes 7} = ZZZZII$$

$$H^{\otimes 7} \bar{Z} H^{\otimes 7} = \bar{X} = X \otimes X \otimes X \otimes X \otimes X \otimes X \otimes X$$

$$CNOT^{\otimes 7} (\bar{Z} \otimes \bar{Z}) CNOT^{\otimes 7} = \bar{I} \otimes \bar{Z}$$

(same for typical $\bar{X} \otimes \bar{X}, \bar{I} \otimes \bar{X}$, etc.)

$$\begin{aligned} &= (iXZ) \otimes (iXZ) \otimes (iXZ) \otimes (iXZ) \otimes I \otimes I \otimes I \\ &= (i^4) (XXXXIIII) (ZZZZIIII) \\ &= (XXXXIIII) (ZZZZIIII) \end{aligned}$$



Transversal gates

Transversal gates naturally have fault-tolerant properties.
Different codes admit different transversal gates.

Example: transversal gates in Steane code.

$$S^{\otimes 7} (\bar{X}) S^{\dagger \otimes 7} = Y^{\otimes 7} = (iXZ)^{\otimes 7} = i^7 \bar{X} \bar{Z} = -i \bar{X} \bar{Z}$$

⇒ use $S^\dagger$ instead!



Transversal gates

Example: non-transversal H in 5-qubit code.

$$\begin{array}{l}
 XZZXI \\
 IXZZX \\
 XIZZZ \\
 ZYIXZ
 \end{array}
 \xrightarrow{H^{\otimes 5}}
 \begin{array}{l}
 ZXXZI \\
 XZZXI \\
 XIZZZ \\
 ZYIXZ
 \end{array}$$

Handwritten calculation for the first two rows:

$$\begin{array}{r}
 ZXXIXZ \\
 IXZZX \\
 \hline
 ZIZY \neq ZXXZI
 \end{array}$$

Handwritten calculation for the last two rows:

$$\begin{array}{r}
 ZXXIXZ \\
 XZZXI \\
 \hline
 ZYIXZ \neq ZXXZI
 \end{array}$$

Example: transversal SH in 5-qubit code.

$\Rightarrow$ try it yourself

$$(SH)^{\otimes 5} \bar{Z} (HS^\dagger)^{\otimes 5} = i \bar{X} \bar{Z} = \bar{Y}$$

Transversal T

Is T transversal in the Steane code? $\Rightarrow$ NO

$ZZZZ III$

$ZZII ZZI$

$ZIZI ZIZ$

$XXXX III$

$XXII XXI$

$XIXI XIY$

$$|1111000\rangle \sim |\bar{0}\rangle$$

$$|0000111\rangle \sim |\bar{1}\rangle$$

$$T|0\rangle = |0\rangle$$

$$T|1\rangle = e^{i\pi/4}|1\rangle$$

$$(T^{\otimes 7})|1111000\rangle = (e^{i\pi/4})^4|1111000\rangle = -|1111000\rangle$$

$$(T^{\otimes 7})|0000111\rangle = e^{3i\pi/4}|0000111\rangle \neq e^{i\pi/4}|0000111\rangle$$

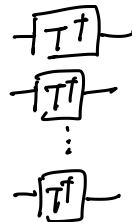
Transversal T

Smallest code with $d = 3$ that admits a transversal T . $[[15, 1, 3]]$
 quantum Reed-Muller Code

$|0\rangle$: weight 8
 $|1\rangle$: weight 7

Z	Z	Z	Z	I	I	I	I	I	I	I	I	I	I	I
Z	Z	I	I	Z	Z	I	I	I	I	I	I	I	I	I
Z	I	Z	I	Z	I	Z	I	I	I	I	I	I	I	I
Z	Z	I	I	I	I	I	I	Z	Z	I	I	I	I	I
Z	I	Z	I	I	I	I	I	Z	I	Z	I	I	I	I
Z	I	I	I	Z	I	I	I	Z	I	I	I	Z	I	I
Z	Z	Z	Z	Z	Z	Z	Z	I	I	I	I	I	I	I
Z	Z	Z	Z	I	I	I	I	Z	Z	Z	Z	I	I	I
Z	Z	I	I	Z	Z	I	I	Z	Z	I	I	Z	Z	I
Z	I	Z	I	Z	I	Z	I	Z	I	Z	I	Z	I	Z
X	X	X	X	X	X	X	X	I	I	I	I	I	I	I
X	X	X	X	I	I	I	I	X	X	X	X	I	I	I
X	X	I	I	X	X	I	I	X	X	I	I	X	X	I
X	I	X	I	X	I	X	I	X	I	X	I	X	I	X

Table 11.1: The stabilizer for the 15-qubit code.



Q: what can we say about H and $CNOT$?

Screenshot: Gottesman, Chapter 11

NO

YES (transversal CNOT)

The Eastin-Knill theorem

Source: Gottesman, Theorem 11.7

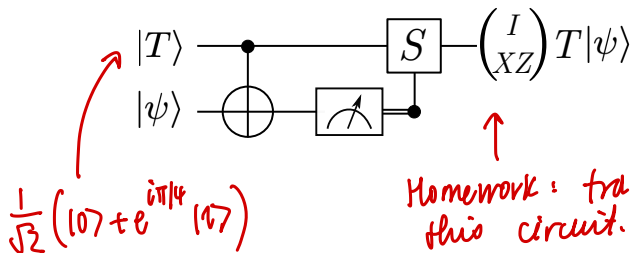
“If Q is an $((n, K, d))_q$ QECC with $d \geq 2$, and G is the set of logical unitary gates that have transversal gadgets for Q , then G is a discrete group. In particular, G is not dense in $SU(K)$.”

If a transversal implementation isn't available, how do we implement the “last” gate fault-tolerantly?

Magic states and gate teleportation

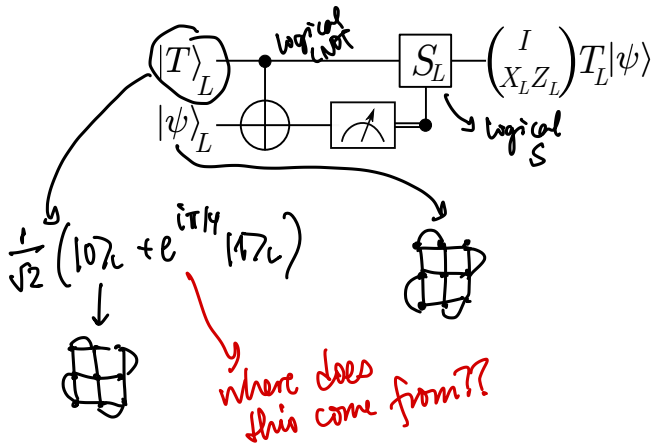
Special resource states (magic states) can be applied with gate teleportation to implement non-Clifford operations.

Analysis at the physical level:



Magic states and gate teleportation

Challenge: this must happen at the *logical* level.



Magic state preparation (unrotated surface code)

Example: prepare a $|\mathcal{T}\rangle_L = \frac{1}{\sqrt{2}} \left(|0\rangle_L + e^{\frac{i\pi}{4}} |1\rangle_L \right)$.

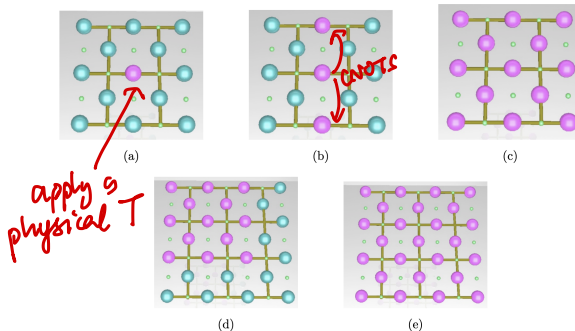
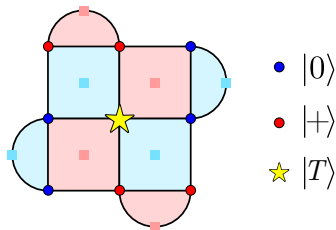


Figure 8. Injecting an arbitrary state $|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$ into a planar surface: a) the blue qubits are prepared in $|0\rangle$, and the pink is the entangled state $|\Psi\rangle$; b) CNOT operations are performed to create $\alpha|000\rangle + \beta|111\rangle$ on the pink qubits; c) measuring stabilizers gives a distance-3 surface in the logical $|\Psi\rangle$ state; d) prepare the blue qubits in $|0\rangle$; e) merging in the blue qubits gives a distance-4 surface in $|\Psi\rangle$.

Image: Horsman, Fowler, Devitt, and van Meter (2012), *Surface code quantum computing by lattice surgery*, NJP

Magic state preparation (rotated surface code)

Example: prepare a $|T\rangle_L = \frac{1}{\sqrt{2}} \left(|0\rangle_L + e^{\frac{i\pi}{4}} |1\rangle_L \right)$.

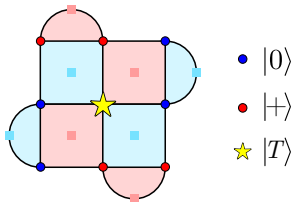


For more information: Lingling Lao and Ben Criger. 2022. Magic state injection on the rotated surface code. In Proceedings of the 19th ACM International Conference on Computing Frontiers (CF '22).

Magic state preparation (rotated surface code)

Analysis in Lao and Criger assumes circuit-level noise:

- Pauli errors after one-(two-)qubit gates, $p = p_1/3$ ($p_2/15$)
- initialization error p_{IN} and measurement error p_M



*Supplementary
→ assignment.*

Overall logical error rate given as

$$\frac{3}{5} p_2 + p_{IN} + \frac{2}{3} p_1 + O(p^2)$$

Magic state distillation

What if the magic states aren't good enough? *Distill* better ones.

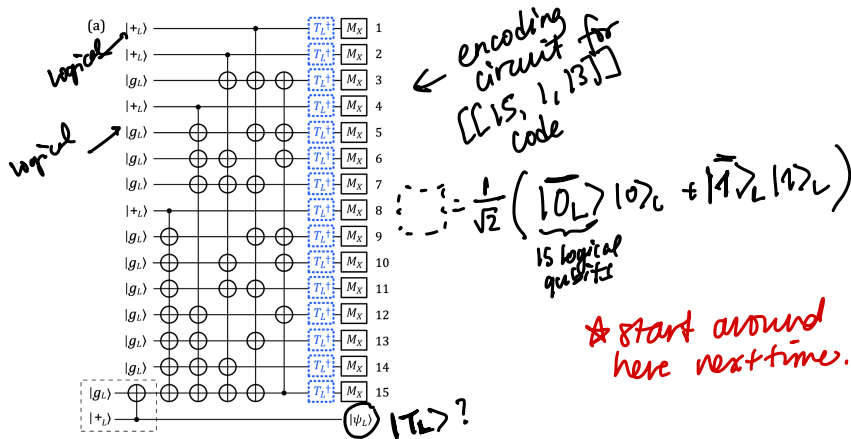
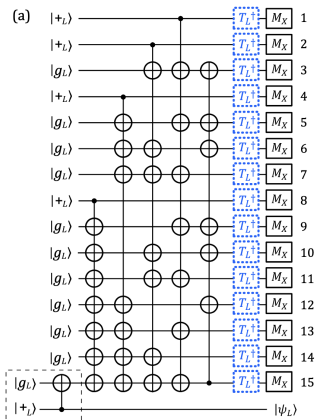


Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

Magic state distillation

What if the magic states aren't good enough? *Distill* better ones.



Measure and proceed if all $+1$:

$$X_{S1} = X_4 X_5 \cdots X_{10} X_{11}$$

$$X_{S2} = X_1 X_2 X_7 X_{15}$$

$$X_{S3} = X_2 X_5 X_{10} X_{13}$$

$$X_{S4} = X_1 X_2 X_5 X_6 X_9 X_{10} X_{13} X_{14}$$

If circuit is perfect and error rate of T is p ,

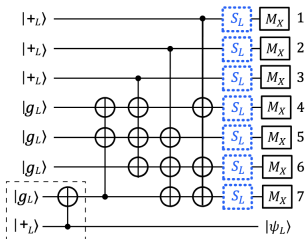
- success probability $1 - 15p$
- logical error probability $35p^3$

Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

Magic state distillation

Easier to understand with this distillation circuit:

Measure and proceed if all +1:



If circuit is perfect and error rate of T is p ,

- success probability
- logical error probability

Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

Magic state distillation

Easier to understand with this distillation circuit:

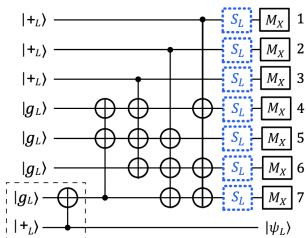


Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

Magic state distillation

Can perform a similar analysis for the $[[15, 1, 3]]$ code.

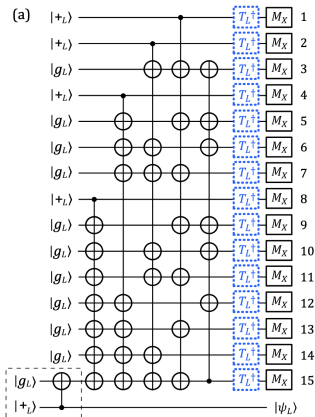


Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

Magic state distillation

If error rate isn't good enough, can concatenate:

- first level error probability $35p^3$
- second level error probability $35(35p^3)^3 \sim O(p^9)$
- ...

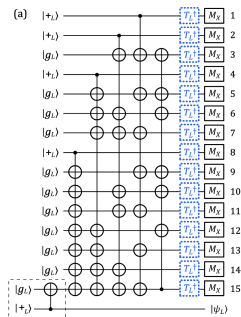
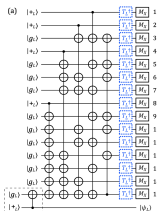


Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

Next time

Next class: final lecture

- Fault-tolerant resource estimation

Action items:

- ① Prepare exam reference sheet

Recommended reading for this / next class:

- Gottesman Chs. 11 and 13; N & C 10.6
- B. Eastin and E. Knill (2009). "Restrictions on Transversal Encoded Quantum Gate Sets". Physical Review Letters. 102 (11) 110502.
- [PennyLane tutorial on magic states](#)
- [PennyLane tutorial on state distillation](#)